

Lesson 7 Quick Reference — Terminal & Errors

Syntax only. For the *why*, see the Lesson 7 reference sheet.

Terminal

Ctrl + ` opens it. Ours runs Git Bash.

```
pwd                # where am I?
ls                 # what is in this folder?
cd cs100           # go down into a folder
cd ..              # go up one level
cd ~               # go home
clear              # wipe the screen
python file.py     # run it (python3 on Mac)
```

Press **Tab** to autocomplete any name — faster, and it can't misspell.

	Means
	right here
	the folder above

Same `.` as `git add .` — "this folder and everything in it."

Paths

Right-click a file → **Copy Relative Path**

```
python unit1/lesson7_sandbox.py
```

- **Relative** — from where you're standing
- **Absolute** — full path from the top of the drive

The error everyone hits

```
python: can't open file 'card.py':  
[Errno 2] No such file or directory
```

Your file is fine. You're in the wrong folder.
`pwd` → `ls` → `cd` into the right one.

Reading a traceback

```
Traceback (most recent call last):  
  File "pizza.py", line 7, in <module>  
    people = int(input("How many? "))  
ValueError: invalid literal for int() with base 10: 'five'
```

Read it **bottom-up**.
Last line = what went wrong. Line number = where to look.

Python stops at the **first** error. Fix one, run again, the next appears.

Three kinds of error

Kind	What happens
Syntax	won't run at all
Runtime	starts, then crashes partway
Logic	runs fine, wrong answer — nothing warns you

Error names

Error	It means
<code>NameError</code>	a name that doesn't exist — usually a typo
<code>TypeError</code>	wrong kind of thing (<code>"5" + 3</code>)
<code>ValueError</code>	right kind, bad content (<code>int("3.7")</code>)
<code>SyntaxError</code>	Python can't even read the line
<code>IndentationError</code>	wrong spaces at the start of a line
<code>ZeroDivisionError</code>	divided by zero

Newer Python adds `Did you mean: 'team_name'?` — read the whole last line.

Before you push

ran it **from the terminal** · no traceback · **output read and checked** · header comment · **pushed**